VIDYAPEETH

BATCH CODE: 12-AN-181-MA-2026

SUBJECT NAME: PHYSICS

CHAPTER NAME:

UNIT & MEASUREMENT

Lecture No. 15

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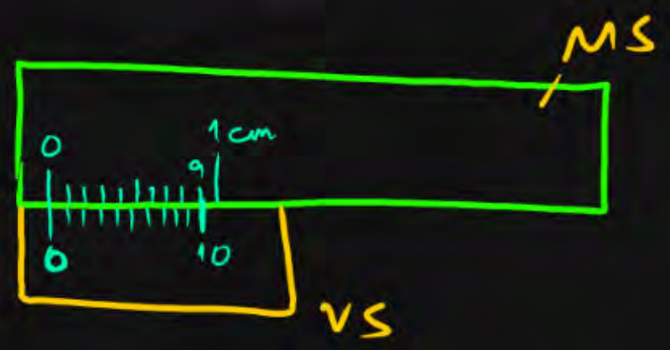
Today's Goal

Subtopic .. Vernier Calliper

Subtopic ..

Subtopic ..

VERNIER CALLIPER



LEAST Count $LC = 1MSD - 1VSD$

$= 1mm - 0.9mm$

$$\boxed{LC = 0.1\,mm}$$

ORDINARY VC

- $1MSD = 1mm$
- $\boxed{10VSD = 9MSD}$
- $1VSD = \frac{9}{10} MSD$

$1VSD = 0.9\,MSD$

$= 0.9\,(1mm)$

$$\boxed{1VSD = 0.9mm}$$

Any General given VC

$n\ VSD = m\ MSD$

$$\boxed{1\ VSD = \frac{m}{n}\ MSD}$$

If $1\ MSD = 1mm$

$$1\ VSD = \frac{m}{n}(1mm)$$

VSD	MSD
Vernier Scale Divisions	Main Scale Division

$1\ MSD = 1mm$

$n > m$

$LC = 1\ MSD - 1\ VSD$

$= 1 - \frac{m}{n}$

$$\boxed{LC = \frac{n-m}{n}}$$

Also :- Special case if VSD = n & MSD = n-1 [ORDINARY]

$$n \text{ VSD} = (n-1)\text{ MSD}$$

$$1 \text{ VSD} = \frac{n-1}{n} \text{ MSD}$$

$$= \frac{n}{n} - \frac{1}{n} \text{ MSD}$$

$$\boxed{1 \text{ VSD} = \left(1 - \frac{1}{n}\right) \text{MSD}}$$

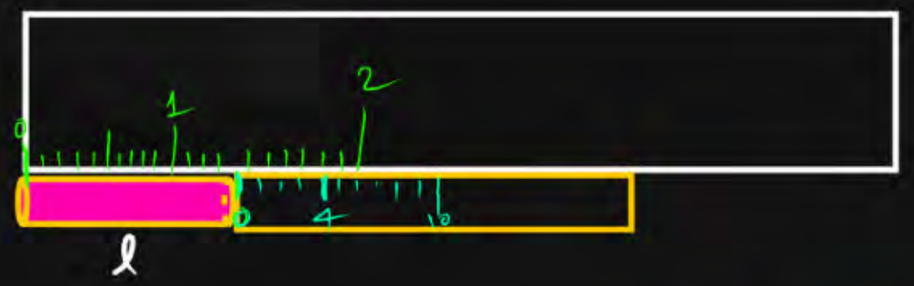
$$LC = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= \left[1 - \left(1 - \frac{1}{n}\right)\right] \text{MSD}$$

$$= \left[1 - 1 + \frac{1}{n}\right] \text{MSD}$$

$$\boxed{LC = \frac{1}{n} \text{ MSD}}$$

$n \rightarrow$ no of VSD



$l = 13 + 4(0.1)$

$= 13.4\ mm$

$\boxed{l = 1.34\ cm}$

If ordinary then LC = 0.1 mm

$l = MSR + n(LC)$

MSR → 1.3 cm or 13 mm

n → n^{th} division of VS coinciding with any of the MS

Q

20 VSD = 16 MSD ; 1 MSD = 1 mm

1 VSD = $\frac{16}{20}$ MSD

1 VSD = $\frac{16}{20}$ mm

LC = 1 MSD - 1 VSD

= 1 mm - $\frac{16}{20}$ mm

= $\frac{20-16}{20} = \frac{4}{20} = \frac{1}{5} = 0.2$ mm

$\frac{n-m}{n} = \frac{20-16}{20}$

$= \frac{4}{20} = \frac{1}{5} = 0.2$

$\frac{1}{n}$ MSD $= \frac{1}{20} \cdot 1$

★★★★★

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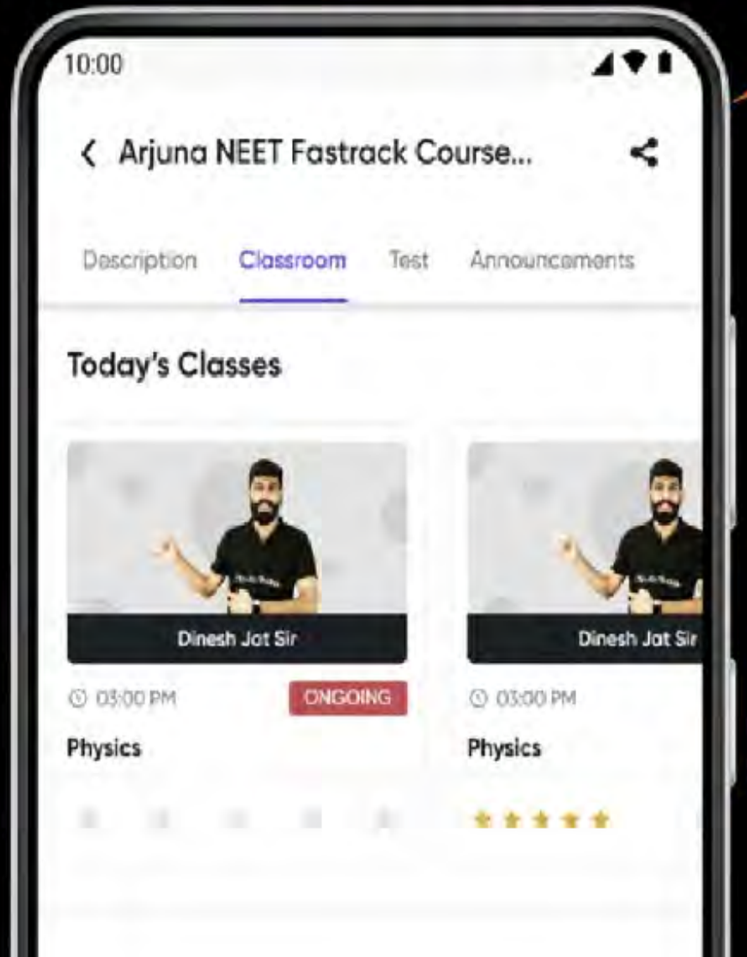
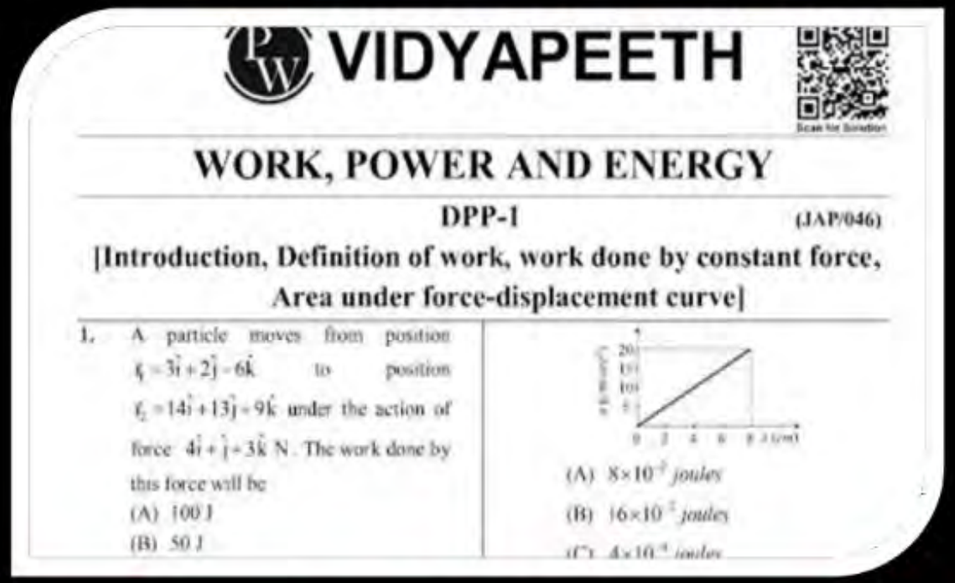


Solve the DPP and check Solution



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